8.2 Span Tables

Maximum allowable spans for No. 2 grade dimensional lumber, from the 2021 International Residential Code. Values are maximums — your plans and local amendments govern.

Before you use these tables

- Values assume No. 2 grade lumber, single span, bearing at each end. Spans are in feet-inches.
- Your local jurisdiction may amend these values. The approved plans and your building department always govern.
- Engineered lumber (I-joists, LVL) has different span capabilities — use the manufacturer's tables.

FLOOR JOISTS — Living Areas

40 psf live load, 10 psf dead load, deflection limit L/360. IRC Table R502.3.1(2).

Floor Joists — 2x8

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	13'-6"	11'-10"	10'-10"	9'-8"
Douglas Fir-Larch	14'-2"	12'-9"	11'-8"	10'-5"
Hem-Fir	13'-2"	12'-0"	11'-3"	10'-2"
Spruce-Pine-Fir	13'-6"	12'-3"	11'-8"	10'-5"

Floor Joists — 2x10

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	16'-2"	14'-0"	12'-10"	11'-5"
Douglas Fir-Larch	18'-0"	15'-7"	14'-3"	12'-9"
Hem-Fir	16'-10"	15'-2"	13'-10"	12'-5"
Spruce-Pine-Fir	17'-3"	15'-5"	14'-1"	12'-7"

Floor Joists — 2x12

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	19'-1"	16'-6"	15'-1"	13'-6"

Floor Joists — 2x12

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Douglas Fir-Larch	20'-11"	18'-1"	16'-6"	14'-9"
Hem-Fir	20'-4"	17'-7"	16'-1"	14'-4"
Spruce-Pine-Fir	20'-7"	17'-10"	16'-3"	14'-7"

Notes: O.C. = on center. For cantilevers, maximum overhang = 1/4 of the back span. Provide blocking or bridging at mid-span for joists over 8 feet.

CEILING JOISTS — Attic With Limited Storage

20 psf live load (uninhabitable attic with limited storage), deflection limit L/240. IRC Table R802.4.1(2). If your attic will never carry storage, code allows longer spans — these values are the safe, conservative choice.

Ceiling Joists — 2x6

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	13'-11"	12'-0"	11'-0"	9'-10"
Douglas Fir-Larch	15'-0"	13'-0"	11'-11"	10'-8"
Hem-Fir	14'-5"	12'-8"	11'-7"	10'-4"
Spruce-Pine-Fir	14'-9"	12'-10"	11'-9"	10'-6"

Ceiling Joists — 2x8

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	17'-7"	15'-3"	13'-11"	12'-6"
Douglas Fir-Larch	19'-1"	16'-6"	15'-1"	13'-6"
Hem-Fir	18'-6"	16'-0"	14'-8"	13'-1"
Spruce-Pine-Fir	18'-9"	16'-3"	14'-10"	13'-3"

Ceiling Joists — 2x10

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	20'-11"	18'-1"	16'-6"	14'-9"
Douglas Fir-Larch	23'-3"	20'-2"	18'-5"	16'-5"
Hem-Fir	22'-7"	19'-7"	17'-10"	16'-0"
Spruce-Pine-Fir	22'-11"	19'-10"	18'-2"	16'-3"

COMMON RAFTERS — Snow Load Areas

30 psf ground snow load, 10 psf dead load, ceiling not attached to rafters, deflection limit L/180. IRC 2021 Table R802.4.1(3). Higher snow areas require shorter spans — check your local ground snow load.

Rafters — 2x6

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	12'-11"	11'-2"	10'-2"	9'-2"
Douglas Fir-Larch	14'-0"	12'-1"	11'-0"	9'-10"
Hem-Fir	13'-7"	11'-9"	10'-9"	9'-7"
Spruce-Pine-Fir	13'-9"	11'-11"	10'-11"	9'-9"

Rafters — 2x8

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	16'-4"	14'-2"	12'-11"	11'-7"
Douglas Fir-Larch	17'-8"	15'-4"	14'-0"	12'-6"
Hem-Fir	17'-2"	14'-11"	13'-7"	12'-2"
Spruce-Pine-Fir	17'-5"	15'-1"	13'-9"	12'-4"

Rafters — 2x10

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	19'-5"	16'-10"	15'-4"	13'-9"
Douglas Fir-Larch	21'-7"	18'-9"	17'-1"	15'-3"
Hem-Fir	21'-0"	18'-2"	16'-7"	14'-10"
Spruce-Pine-Fir	21'-4"	18'-5"	16'-10"	15'-1"

Rafters — 2x12

Species (No. 2 grade)	12" O.C.	16" O.C.	19.2" O.C.	24" O.C.
Southern Pine	22'-10"	19'-10"	18'-1"	16'-2"
Douglas Fir-Larch	25'-1"	21'-8"	19'-10"	17'-9"
Hem-Fir	24'-4"	21'-1"	19'-3"	17'-3"
Spruce-Pine-Fir	24'-8"	21'-5"	19'-6"	17'-6"

HEADERS & GIRDERS — Exterior Bearing Walls

Douglas Fir-Larch, Hem-Fir, Southern Pine, or Spruce-Pine-Fir, No. 2 or better. 30 psf ground snow, 24-foot building width. IRC 2021 Table R602.7(1). "Jacks" = number of jack (trimmer) studs required at EACH end of the header.

Header Spans (span + jack studs per end)

Header Size	Supporting Roof + Ceiling	+ One Floor	+ Two Floors
2-2x4	3'-1" (1 jack)	2'-6" (1 jack)	2'-1" (1 jack)
2-2x6	4'-7" (1 jack)	3'-9" (1 jack)	3'-2" (2 jacks)
2-2x8	5'-9" (1 jack)	4'-10" (2 jacks)	4'-0" (2 jacks)
2-2x10	6'-10" (2 jacks)	5'-8" (2 jacks)	4'-9" (2 jacks)
2-2x12	8'-1" (2 jacks)	6'-8" (2 jacks)	5'-7" (2 jacks)
3-2x8	7'-3" (1 jack)	6'-0" (1 jack)	5'-0" (2 jacks)
3-2x10	8'-7" (1 jack)	7'-2" (2 jacks)	5'-11" (2 jacks)
3-2x12	10'-1" (2 jacks)	8'-5" (2 jacks)	7'-0" (2 jacks)
4-2x8	8'-4" (1 jack)	6'-11" (1 jack)	5'-9" (1 jack)
4-2x10	9'-11" (1 jack)	8'-3" (2 jacks)	6'-10" (2 jacks)
4-2x12	11'-8" (1 jack)	9'-8" (2 jacks)	8'-1" (2 jacks)

Notes: "+ One Floor" / "+ Two Floors" = header also supports one or two center-bearing floors above. Install headers on edge, crown up. King stud required each side in addition to jack studs. Wider buildings, higher snow loads, or clear-span floors above require larger headers — see IRC Table R602.7(1) or consult an engineer.

DECK JOISTS

40 psf live load, 10 psf dead load, wet service. Span measured from face of support to face of support, no overhang. IRC 2021 Table R507.6.

Deck Joists — Southern Pine #2

Joist Size	**12" O.C.**	**16" O.C.**	**24" O.C.**
2x6	9'-11"	9'-0"	7'-7"
2x8	13'-1"	11'-10"	9'-8"
2x10	16'-2"	14'-0"	11'-5"
2x12	18'-0"	16'-6"	13'-4"

Deck Joists — Doug Fir-Larch / Hem-Fir / SPF #2

Joist Size	12" O.C.	16" O.C.	24" O.C.
2x6	8'-10"	8'-0"	6'-10"
2x8	11'-8"	10'-7"	8'-8"
2x10	14'-11"	13'-0"	10'-7"
2x12	17'-5"	15'-1"	12'-4"

Deck notes: use pressure-treated or naturally durable lumber; joist hangers at ledger connections; cantilever (overhang) limited to 1/4 of the actual joist span; check local amendments.

IMPORTANT DISCLAIMER

These tables are general reference for common residential conditions, transcribed from the 2021 International Residential Code. Actual requirements vary with species, grade, moisture, load duration, snow load, and local amendments. Confirm all structural sizing with your building department and, where required, a licensed engineer.